

A novel machine learning algorithm selects proteome signature to specifically identify cancer exosomes

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Abstract

Non-invasive early cancer diagnosis remains challenging due to the low sensitivity and specificity of current diagnostic approaches. Exosomes are membrane-bound nanovesicles secreted by all cells that contain DNA, RNA, and proteins that are representative of the parent cells. This property, along with the abundance of exosomes in biological fluids makes them compelling candidates as biomarkers. However, a rapid and flexible exosome-based diagnostic method to distinguish human cancers across cancer types in diverse biological fluids is yet to be defined. Here, we describe a novel machine learning-based computational method to distinguish cancers using a panel of proteins associated with exosomes. Employing datasets of exosome proteins from human cell lines, tissue, plasma, serum and urine samples from a variety of cancers, we identify Clathrin Heavy Chain (CLTC), Ezrin, (EZR), Talin-1 (TLN1), Adenylyl cyclase-associated protein 1 (CAP1) and Moesin (MSN) as highly abundant universal biomarkers for exosomes and define three panels of pan-cancer exosome proteins that distinguish cancer exosomes from other exosomes and aid in classifying cancer subtypes employing random forest models. All the models using proteins from plasma, serum, or urine-derived exosomes yield AUROC scores higher than 0.91 and demonstrate superior performance compared to Support Vector Machine, K Nearest Neighbor Classifier and Gaussian Naive Bayes. This study provides a reliable protein biomarker signature associated with cancer exosomes with scalable machine learning capability for a sensitive and specific non-invasive method of cancer diagnosis.

eLife assessment

This **important** study introduces a novel AI method for the analysis of published data, with practical implications for early cancer diagnosis. The results are supported by **compelling** evidence.

Introduction

Tissue biopsies have traditionally been a definitive way to diagnose and stage cancer, however, a biopsy may not be easily accessible for many tumors such as those in the pancreas, lung and brain^{1,2,3,4,5}. In addition, the small amount of biopsied tissue does not represent the entire heterogeneous pathological profile of the tumor⁶. In recent years, liquid biopsy has emerged as a plausible diagnostic and monitoring approach with the capability to detect tumor biomarkers in more accessible biological fluids such as plasma, serum and urine⁷. Detectable tumor biomarkers can include circulating tumor DNA (ctDNA), circulating tumor cells (CTCs) and exosomes⁸.

Exosomes are extracellular vesicles of endosomal origin that are between ~40-180 nm in diameter and have been shown to mediate intercellular communication in health and disease^{8,9,10}. They can contain a variety of biomolecules including DNA, RNA, proteins, lipids, metabolites and other materials representative of the parent cell⁸. Exosomes are present at high concentrations in biological fluids, which is a potential advantage as a biomarker^{11,12}. Exosomal mRNAs and miRNAs have been investigated intensively as diagnostic biomarkers, and mounting evidence suggests that exosomal proteins circulating in biological fluids could be used for cancer diagnosis and monitoring cancer progression^{7,13}. Challenges that remain include standardization of methods for consistent exosome isolation from various tissues, identification of biomarkers that distinguish cancer and normal exosomes across different cancer types, and the identification of biomarkers that are unique to specific biological fluids (e.g., plasma, serum, urine, etc.).

In addition to accurately identifying exosomal protein biomarkers, it is also challenging to successfully utilize them to diagnose cancers due to intratumor and interpatient heterogeneity¹⁴. Conventional diagnostic approaches predominantly rely on a single biomarker, which is often not specific or sensitive^{15,16}. However, recent advances in machine learning algorithms connected to artificial intelligence provide an opportunity to construct a classifier that identifies a panel of exosomal protein biomarkers that would possess a more comprehensive ability to reflect the complex disease status of different patients and distinguish cancer samples from normal samples with significantly improved sensitivity and specificity.

Results

Unbiased proteomics analysis of exosomes identifies 18 abundant plasma membrane protein markers for various human cell lines

To identify universal exosomal protein biomarkers for differentiating cancer from non-cancer exosomes, we analyzed protein abundance data from 228 cancer and 57 control cell line-derived exosomes, representing various cancer types (**Figure 1A**; **Supplementary Table S1**). Because studies employ distinct isolation and mass spectrometry quantification techniques, the number of identified proteins is different among the studies. To overcome the bias caused by such technical factors, we examined the proteins common to all studies and identified 1124 overlapping proteins (**Figure 2A**). To determine the heterogeneity among cancer and control cell line-derived exosomes, we performed principal component analysis (PCA) using these 1124 proteins in 285 different studies with cancer and control cell line-derived exosomes (**Figure 2B**). The PCA indicated that the exosomes derived from cancer and control cell lines are heterogeneous and show significant variation in protein expression across cell lines.

We next investigated the frequency of the proteins detected in the exosomes from all cell lines, only the cancer cell lines and just the control cell lines. Commonly used exosome biomarkers (e.g., CD9, tetraspanins) were examined first, and from the 12 traditional exosome markers^{17,18,19},

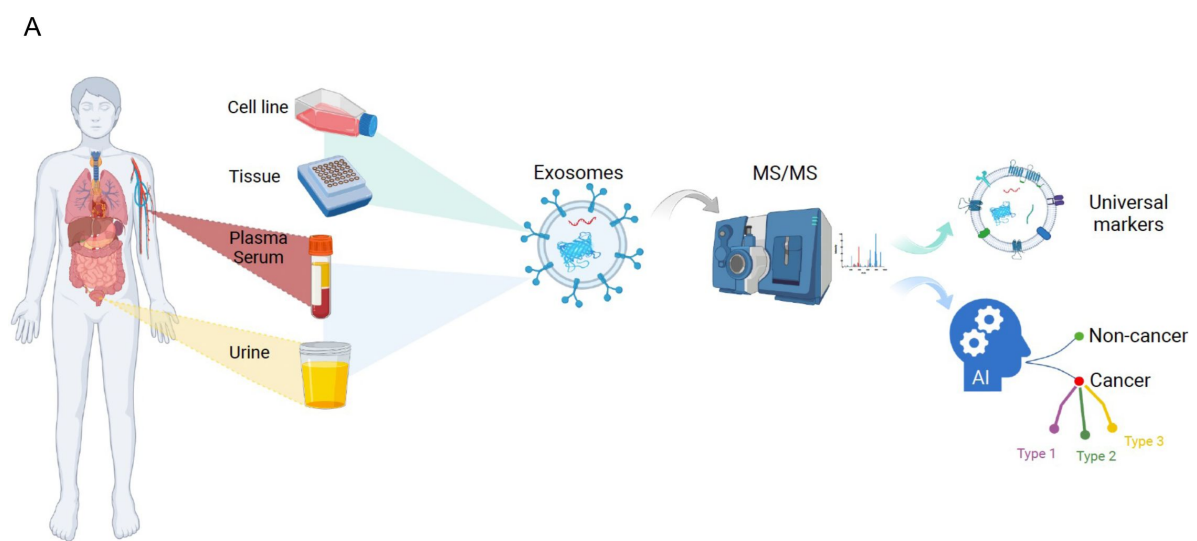


Figure 1.

Overview of the study.

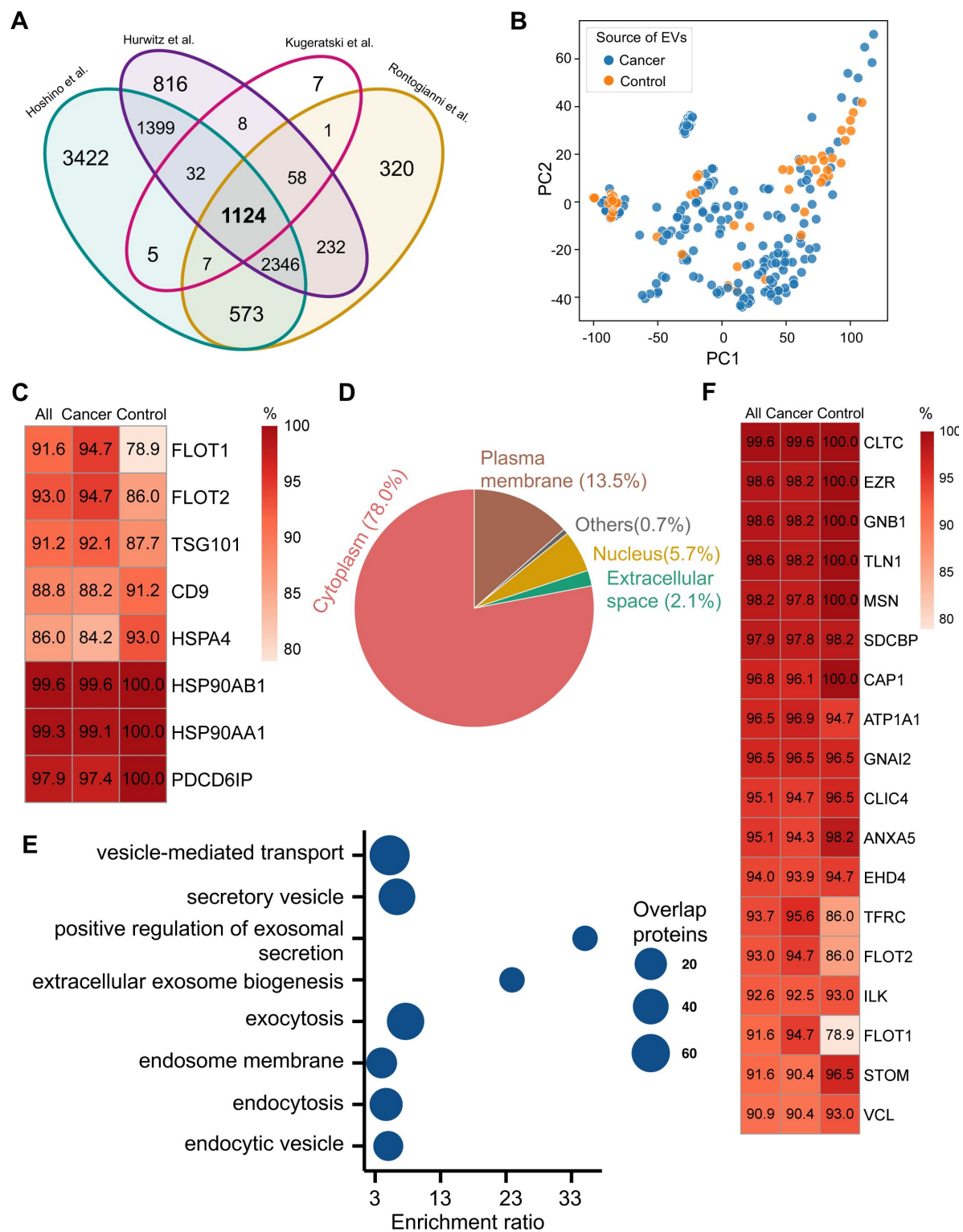


Figure 2.

Proteomic characterization of exosomes derived from 285 cell lines from four studies.

(A) Overlapped proteins from four different studies of cell line-derived exosomes. (B) PCA plot of cancer and control cell line-derived exosomes. (C) Positivity for 8 commonly used exosomal protein biomarkers in various cell lines. The percentage of samples expressing each protein is shown in the boxes. Darker red indicates a higher percentage. (D) Annotation of the proteins detected in more than 90% of all samples. (E) GO and KEGG pathway enrichment analysis of the proteins detected in more than 90% of all samples. (F) Plasma membrane proteins detected in more than 90% of all samples.

only eight proteins were detected among the 1124 overlapping proteins from all cell lines. Six of eight proteins were detected in at least 90% of all samples, with CD9 and HSPA4 detected with the least frequency (**Figure 2C**). In addition, the frequency of FLOT1, FLOT2 and TSG101 proteins was higher in cancer cell line-derived exosomes when compared to control cell line-derived exosomes (**Figure 2C**).

To identify biomarkers detectable at a high frequency in all cancer and control cell line-derived exosomes, we searched for the proteins that were detected in $\geq 90\%$ of all samples (**Supplementary Table S2**). We annotated these proteins using Ingenuity Pathway Analysis (IPA) and found that 78.0% of these proteins localize to the cytoplasm and 13.5% of proteins are associated with the plasma membrane (**Figure 2D**). Gene Ontology (GO) analysis revealed the enrichment of proteins from pathways related to vesicle-mediated transport, secretory vesicles, exocytosis, endocytosis, and other exosome-related pathways (**Figure 2E**). To further explore the utility of the proposed biomarkers, we examined the proteins located on the plasma membrane that met the threshold of detection in $\geq 90\%$ of all samples (**Figure 2F**). Clathrin Heavy Chain (CLTC) was ranked as the top plasma membrane protein detected in 99.6% of all samples and 100% of control samples (**Figure 2F**). In addition, the scaffolding protein, Syntenin-1(SDCBP) was detected at a high frequency of 97.9% of all samples, corroborating previous findings²⁰. Next, we sought to identify unique markers that can identify cancer cell-derived exosomes (cancer exosomes) by filtering out the proteins present in $\leq 10\%$ of 57 control cell line-derived exosomes. Interestingly, Ataxin 2 Like (ATXN2L), which has been reported to promote cancer cell invasiveness and resistance to chemotherapy, was uniquely detected in the cancer cell-derived exosomes (**Figure 3A**)²¹. In total, we identified a set of 18 exosome protein markers that are present at a high abundance in all exosomes examined (**Figure 2F**).

Comparison of exosomal proteins derived from cell lines and tissues identified five universal plasma membrane protein markers

We next sought to investigate common biomarkers for the tissue-derived exosomes across cancer types. We calculated the detection frequency of commonly used exosome markers in the 157 samples (101 cancers; 56 controls). Two established exosome markers, CD63 and TSG101, were only detected in 33.1% and 45.9% of all samples, respectively (**Figure 3B**). To identify high-frequency biomarkers for exosomes derived from both cell lines and tissues, we examined the overlapping proteins that met a threshold of $\geq 90\%$ in all samples for exosomes derived from cell lines and tissues and found 31 common proteins (**Figure 3C**). Among the 31 proteins, there were five proteins that were detected in over 90% of all cell line and tissue-derived exosomes (**Figure 3D**). These include Clathrin Heavy Chain (CLTC), Ezrin, (EZR), Talin-1 (TLN1), Adenylyl cyclase-associated protein 1 (CAP1) and Moesin (MSN).

An exosome proteome signature of 18 proteins can differentiate cancer exosomes from non-cancer exosomes across multiple cancer types

Plasma or serum is the most readily accessible source for non-invasive biopsies. We next sought to identify if exosome proteins in plasma and serum could differentiate cancer exosomes from non-cancer exosomes across multiple cancer types. We pooled exosome proteomics data derived from plasma or serum for 205 cancer and 51 control samples from five different studies^{22–26}, which included breast cancer, colorectal cancer, glioblastoma, lung carcinoma, liver cancer, neuroblastoma, and pancreatic cancer (**Supplementary Table S1**). To account for differences in the methodologies of these studies, we first reviewed the proteins and identified 46 proteins that were detected in all the studies (**Figure 4A**). We then examined their abundances in the 205 cancer and 51 control samples (**Figure 4B**). Although we could detect differences between the

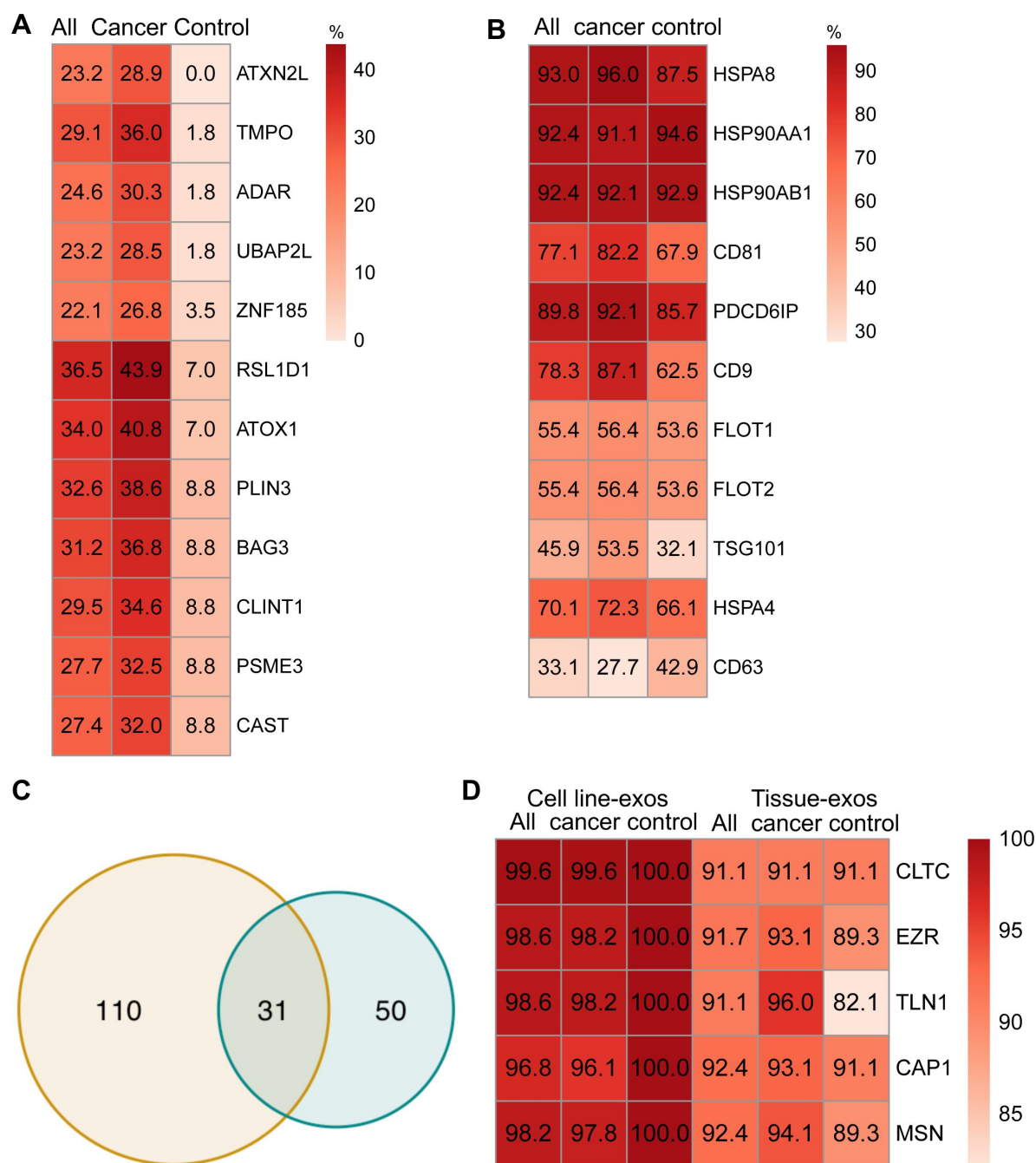


Figure 3.

Proteomic characterization of exosomes derived from cell lines and tissues.

(A) Proteins detected at higher frequency in cancer cell line-derived exosomes. (B) Positivity for 11 commonly used exosomal protein biomarkers in various tissues. (C) Overlapping proteins (>90% frequency) between cell line- and tissue-derived exosomes. (D) Positivity of five plasma membrane proteins detected in more than 90% of both cell line- and tissue-derived exosomes.

exosomes derived from cancer cell and non-cancer cells, it was difficult to classify them robustly based on PCA (**Supplementary Figure S1A**). Therefore, we sought to employ the advanced machine learning algorithm to differentiate cancer exosomes from non-cancer exosomes. We first calculated the mutual information score (MI) for each protein and trained the random forest classifier using the different numbers of top proteins according to their MI score to determine the best set of proteins we should include in the classifier. We found that the model performed best using 18 proteins, where the performance was evaluated by the area under the curve of the receiver operating characteristic curve (AUROC) (**Figure 4C**). Several key cancer-associated proteins were included among these 18 proteins. For example, apolipoprotein C1 (APOC1), which was ranked as the top protein, is significantly decreased in samples of cancer patients (**Supplementary Figure S1B**) and has been previously reported to be down-regulated in the non-small lung cancer, colorectal cancer, papillary thyroid carcinoma and pediatric nephroblastoma^{27–30}.

Employing all 18 proteins with 5-fold cross-validation, we constructed a random forest classifier³¹ to distinguish cancer and control samples and compared it with multiple popular machine learning models, including Support Vector Machine (SVM)³², K Nearest Neighbor Classifier (K-NN)³³ and Gaussian Naive Bayes³⁴. Our random forest classifier demonstrated the highest AUROC (**Figure 4D**). More specifically, it yielded an AUROC of 0.96, with an accuracy of 0.92, a precision of 0.94, and a recall of 0.96 (**Figure 4E & F**). When applied to the independent test set, the model yielded an AUROC of 0.99, an accuracy of 0.95, a precision of 0.96 and a recall of 0.98 (**Figure 4G & H**). Importantly, only one sample was misclassified in the independent test set, and 51 cancer samples were correctly classified. Taken together, our results showed the advantage and clinical potential of applying the random forest classifier model to plasma or serum exosome protein based liquid biopsy for cancer diagnosis.

Five plasma/serum exosomal proteins can reliably differentiate five common cancer types

We next sought to further enhance the clinical utility of the exosomes for differentiating between cancer types. We analyzed proteomics data from plasma or serum derived exosomes from patients with five common cancer types, including breast cancer, colorectal cancer, glioma, lung cancer and pancreatic cancer. Initial PCA revealed differences in exosome levels among cancer patients but failed to distinguish the five cancer types (**Figure 5A**). We determined the crucial features for cancer-type classification by computing mutual information scores for 46 common proteins and built a random forest classifier to determine the optimal number of features to include in the final classifier. We ultimately selected a set of five proteins based on AUROC scores (**Figure 5B**). We then increased the independent testing data size by utilizing 40% of the total samples and used the remaining 60% as the training set to minimize overfitting issues.

Employing the five proteins with 5-fold cross-validation to train the random forest classifier, the model achieved a very high accuracy of 0.99 (**Figure 5C**), and when applied to the independent test set, the model consistently yielded a high accuracy of 0.94 (**Figure 5D**). The abundance of the five protein features varied across the five cancer types, reflecting the potential roles of these proteins in specific cancers (**Figure 5E**). For example, Histidine-rich glycoprotein (HRG), which was abundant in the colorectal cancer plasma-derived exosomes, has been reported to promote the tumor migration of colorectal cancer patients³⁵. Overall, our results demonstrated that this exosome protein-based classification model can reliably differentiate between cancer types and further enhance the diagnostic value of our approach.

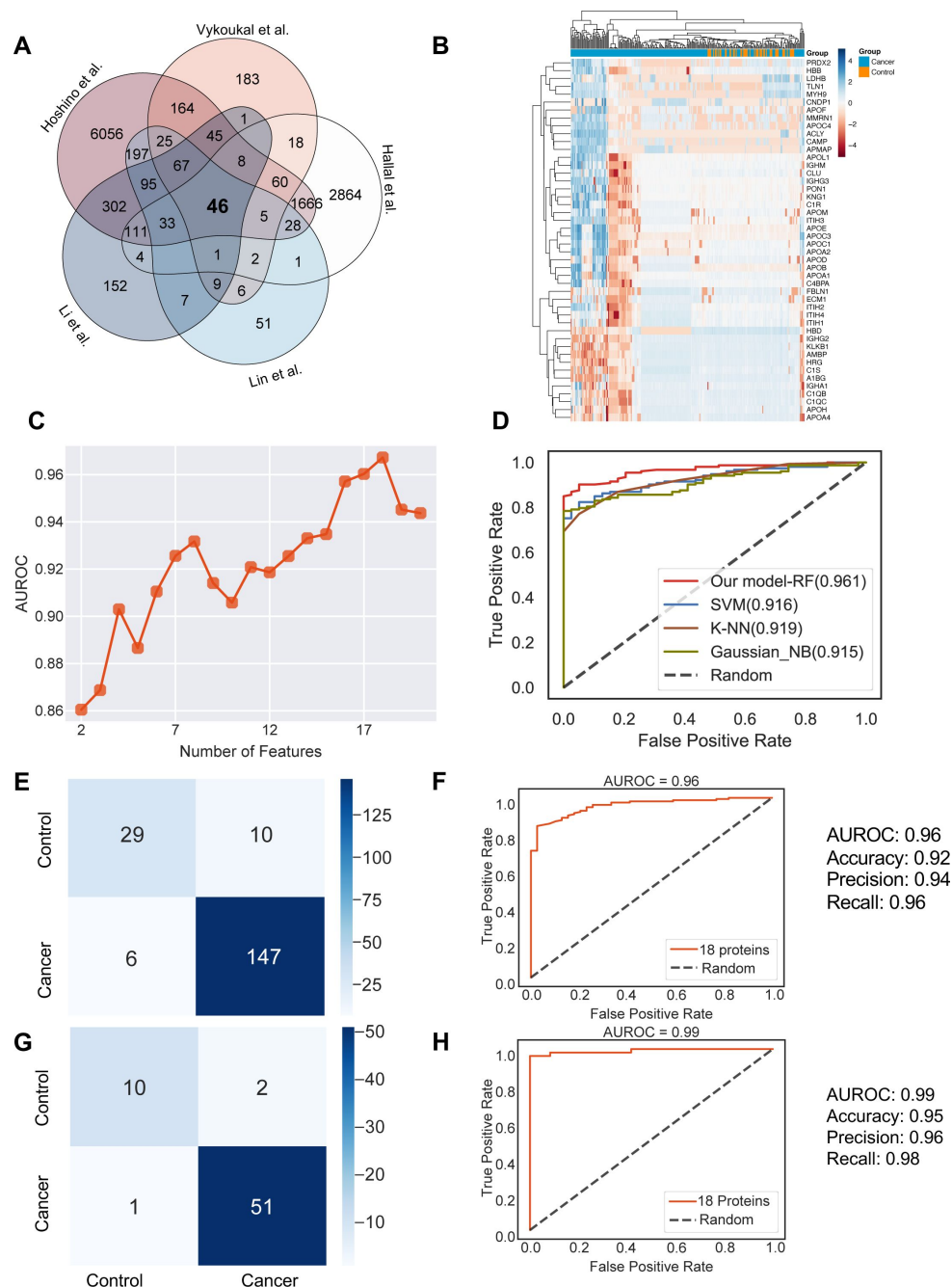


Figure 4.

Identification of the signature proteins of plasma or serum-derived exosomes and the evaluation of random forest classifier.

(A) Overlapping exosome proteins detected in the plasma and serum of 205 cancer and 51 control samples from five studies. (B) Heat map of 46 overlapping exosome proteins in cancer and control plasma or serum samples. (C) AUROC score of the random forest classifier on including various numbers of protein features. (D) AUROC of different models in comparison. (E) Classification error matrix of the 75% training set using a random forest classifier for the 18 selected proteins. The number of samples is indicated in each box. (F) AUROC score of the random forest classifier trained using 75% of the dataset. Other metrics are indicated on right. (G) Classification error matrix of 25% testing set using a random forest classifier for the 18 selected proteins. The number of samples is indicated in each box. (H) AUROC score of the random forest classifier tested using 25% of the dataset. Other metrics are indicated on right.

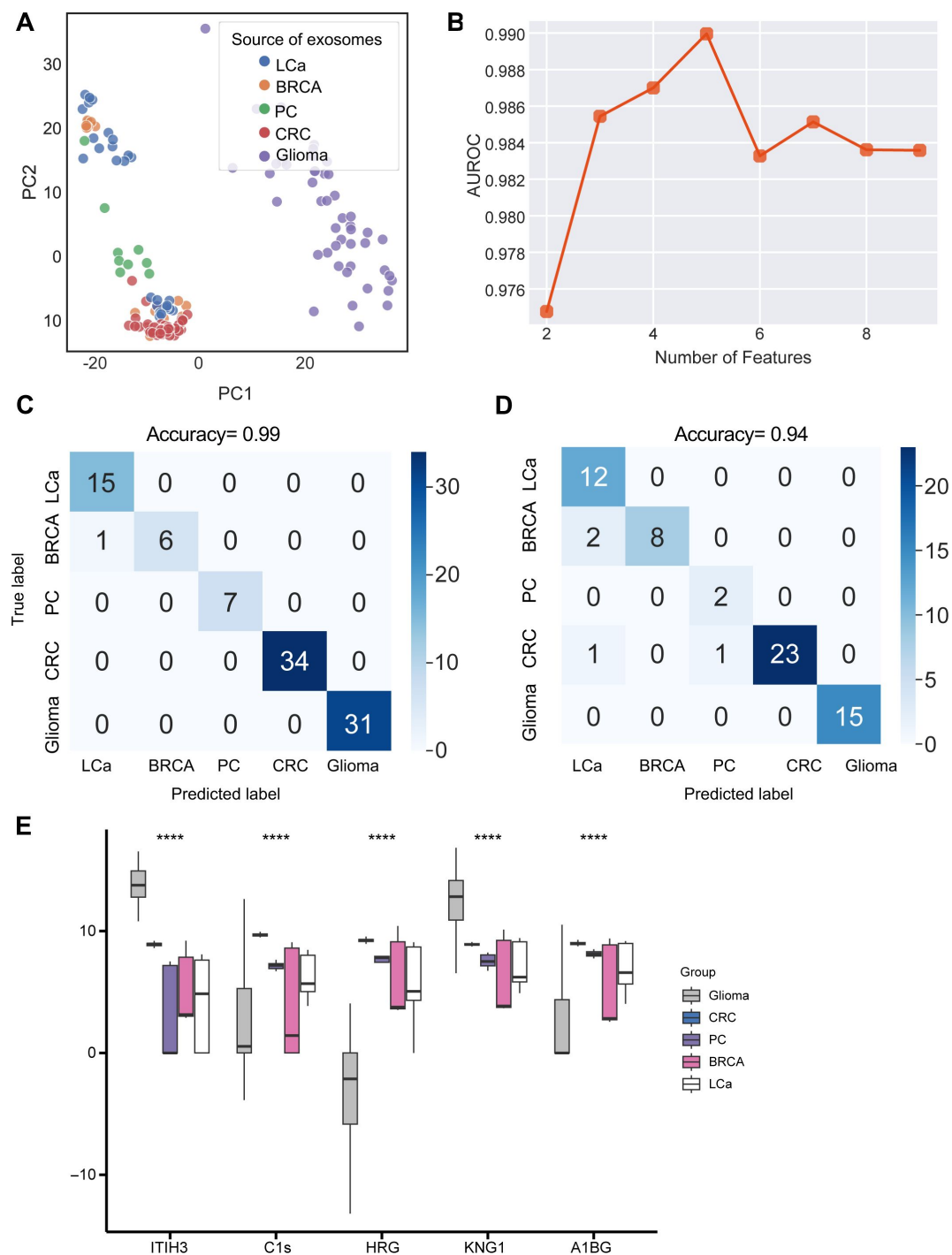


Figure 5.

Identification of signature proteins expressed by plasma or serum-derived exosomes for classifying five common cancer types and evaluation of random forest classifier.

(A) PCA plot of plasma or serum-derived exosomes from five cancer types. (B) AUROC score of the random forest classifier by including various number of protein features. (C-D) Classification error matrix of a 60% training set and 40% testing set to classify the five cancer types using a random forest classifier for the 5 selected proteins. The number of samples is indicated in each box. (E) Protein abundance of five selected protein features in five cancer types.

A urinary exosome proteome signature consisting of 17 proteins detects cancer exosomes across multiple cancer types

Urine is emerging as a superior non-invasive marker for urologic cancers, as its composition directly reflects the physiological changes in the urogenital system³⁶. To test the use of urinary exosome proteins in cancer diagnosis, we collected data from 261 cancer patient samples and 124 control samples from four studies, including bladder cancer, prostate cancer, and renal cancer, lung cancer, cervical cancer, colorectal cancer, esophageal and gastric cancer (**Supplementary Table S1**)^{36–39}. Upon examination of proteins common across all four studies, 229 proteins were identified (**Figure 6A**). PCA revealed the variance between samples but failed to differentiate between cancer and control samples (**Figure 6B**). As described earlier, we next employed the random forest classifier method to differentiate cancer and control samples based on their exosomal proteomic profiles. To reduce the feature space and select the most relevant features, we utilized the mutual information score to rank the 229 protein features and then trained the random forest classifier using varying numbers of the top-ranking proteins (**Figure 6C**). Based on the AUROC scores of including the different number of features, we selected 17 features that resulted in the highest AUROC score (**Figure 6C**). A majority of these 17 proteins displayed significant variations in abundance between cancer and control samples (**Figure 6D**). By training a random forest classifier with the 17 protein features and 5-fold cross-validation, the model achieved an AUROC of 0.96, an accuracy of 0.90, a precision of 0.92, and a recall of 0.93 (**Figure 6E & F**). When tested on an independent set, the model produced an AUROC of 0.91, an accuracy of 0.82, a precision of 0.83, and a recall of 0.92 (**Figure 6G & H**). To summarize, our findings indicated the promising clinical potential of using urinary exosome proteins for the diagnosis of urologic cancers as well as other non-urologic cancer types.

Discussion

Liquid biopsy has numerous benefits in the early detection of cancer, categorizing cancer types, tracking cancer progression, and monitoring response to treatment⁴⁰. Exosomes found in biological fluids can provide a forensic view of their cells of origin. Despite previous studies proposing common protein biomarkers to identify exosomes, a comprehensive set of exosome biomarkers derived from different biological materials has not been established, owing to limitations in isolation and quantification methods^{41–44}. Additionally, a reliable diagnostic tool based on proteins associated with exosomes that can be applied across all cancers is yet to be identified.

Here, we generate a comprehensive proteomics profile of exosomes derived from cell lines, tissues, plasma, serum, and urine from 1083 cancer and control samples. An extensive analysis of these samples showed that several widely used exosome markers, such as CD63, CD81, HSP70, and HSPA8, are absent in exosomes derived from a subset of cell lines. Further, FLOT1, FLOT2, TSG101, and CD63 are present at low levels (< 60%) in all exosomes derived from tissues, indicating a need for the identification of additional universal markers for both cell line- and tissue-derived exosomes. Our study identifies five highly abundant universal exosome biomarkers-CLTC, EZR, TLN1, CAP1, and MSN-that were present in over 90% of all cell line- and tissue-derived samples. Additionally, we found that ATXN2L was only present in cancer exosomes and absent in non-cancer exosomes. In this regard, GPC1 which was identified in many studies identified as cancer exosomes specific marker, was not identified here as many of databases did not pick up this protein in analysis.

Here we describe a novel computational approach using the random forest classifier method to define exosome protein panels that serve as effective biomarkers specifically for plasma, serum or urine across cancer types. By training the random forest model and testing with independent

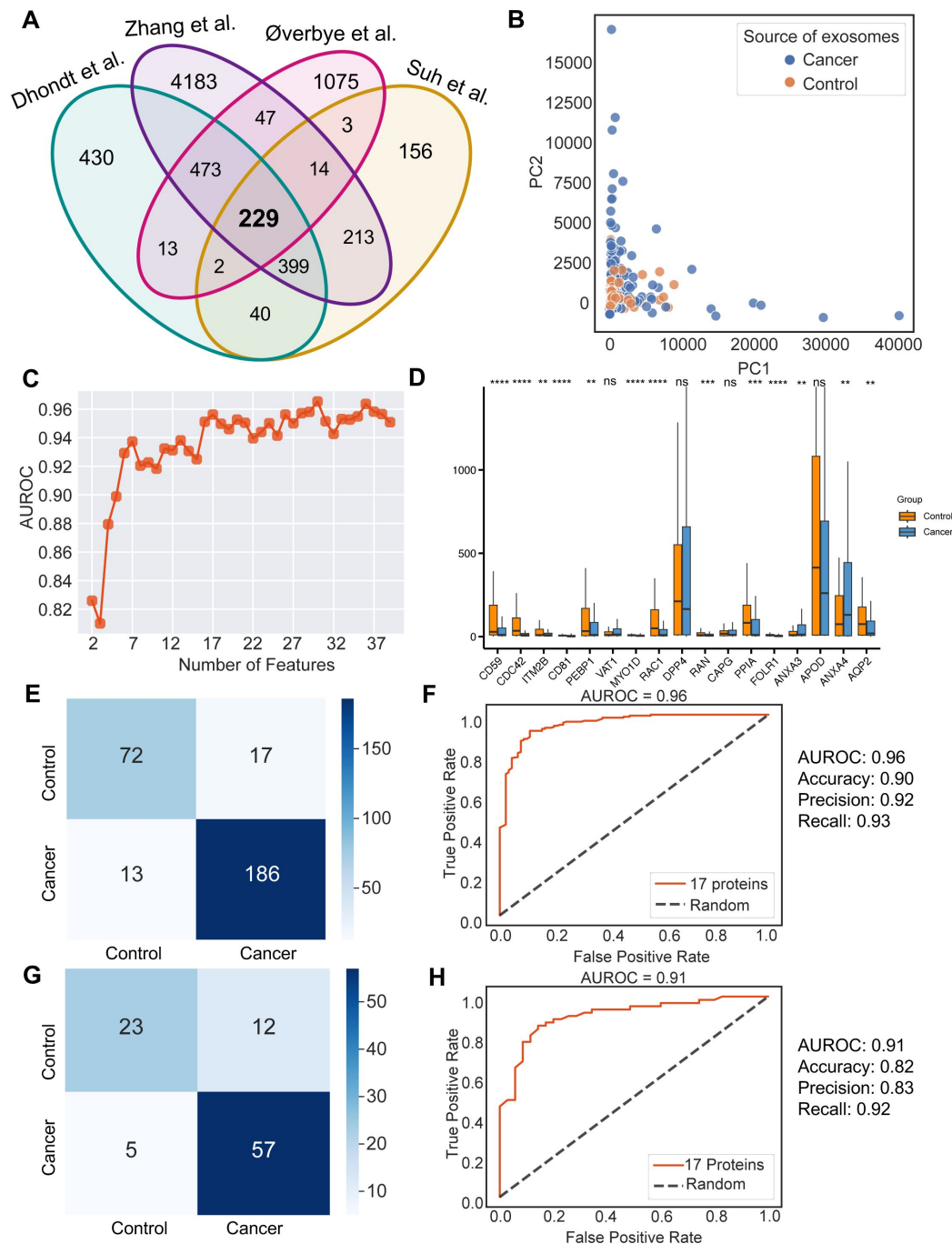


Figure 6.

Identification of signature proteins expressed by urine-derived exosomes and evaluation of random forest classifier.

(A) Overlapping exosome proteins detected in the urine from 261 cancer and 124 control samples from four studies. (B) PCA plot of cancer and control urine-derived exosomes. (C) AUROC score of the random forest classifier by including a various number of protein features. (D) Protein abundance of 17 selected protein features in cancer- and control urine-derived exosomes. (E) Classification error matrix of 75% training set using a random forest classifier for the 17 selected proteins. The number of samples is indicated in each box. (F) AUROC score of the random forest classifier trained using 75% of the dataset. Other metrics are indicated on right. (G) Classification error matrix of 25% testing set using a random forest classifier for the 17 selected proteins. The number of samples is indicated in each box. (H) AUROC score of the random forest classifier tested using 25% of the dataset. Other metrics are indicated on right.

datasets, our model yields excellent scores in AUROC, sensitivity, and specificity and differentiates cancer exosomes from non-cancer exosomes. We show that this approach can also be used to classify, with high accuracy, five common cancer types based on their exosome protein signatures.

A majority of the protein makers identified in this study have demonstrable biological relevance in cancer. As an example, ITIH3, which was identified among the protein features for the plasma or serum-based classifier and highly abundant in cancer samples, was reported to be more highly expressed in the plasma of gastric cancer samples compared to the control⁴⁵ and increased with tumor staging of clear cell renal cell carcinoma patients⁴⁶. Importantly, the biomarker panels identified for cell lines, tissues, plasma/serum, and urine overcome the bias associated with the isolation and quantification method as well as the inter-patient variability inherent to the complex process of cancer. These panels are designed to be applied to a variety of cancers.

Collectively, our results demonstrate that exosome protein features can be utilized as reliable biomarkers for the early detection of cancer, classification of cancer types, and potentially for diagnosing tumors of undetermined origin. These results have the potential to advance the development and standardization of innovative and optimized methods for the isolation of exosomes and the implementation of routine plasma, serum- and urine-based exosome screening in clinical settings.

Methods

Public exosome proteomics data

We collected publicly available exosome protein data from cell lines, plasma, serum, and urine from previous studies as summarized in **Supplementary Table S1**^{20, 22–26, 36–39, 43, 47}. We obtained the raw spectral count and intensity data from the original paper or directly from the authors through communication. The obtained data was log normalized accordingly.

Feature selection and machine learning algorithm

We employed mutual information scores to evaluate the importance of protein features for each prediction, quantifying the amount of uncertainty reduction for one variable given the knowledge of another variable. We calculated the mutual information score for each protein with a target label (cancer/control) using SelectKBest from the scikit-learn 1.1.1 Python library.

The mutual information score quantifies the decrease in uncertainty of one variable when the value of the other variable is known. We performed the mutual information score calculation twenty times and computed the average score for each protein as the ultimate score. According to the sample size and characteristics of the data collected for plasma and urine-derived exosomes, we selected a customized number of best features for each prediction. To select the optimum number of proteins to include in the prediction, we ranked the protein features based on the mutual information score and built a random forest classifier to evaluate performance on including a range of number of features. The random forest model can decrease the probability of over-fitting and enhance the resilience towards outliers and input data noise. The area under the curve of the receiver operating characteristic curve (AUROC) was employed to evaluate the performance of the classifier. We also calculated accuracy, precision and recall for comprehensive evaluation. All models were evaluated using 5-fold cross-validation with stratified train-test splits that preserved the percentage of samples for the prediction target. We also tested the performance of alternate machine learning algorithms including support vector classifier, K nearest neighbor classifier and Gaussian Naive Bayes. Overall, the random forest classifier achieved the best performance in our analysis (**Supplementary Figure S1B**). To visualize high-dimensional

datasets, kernel principal component analysis (PCA) from scikit-learn 1.1.1 Python library was employed to perform PCA and plots were generated using the scatterplot function from seaborn 0.11.2 Python library.

Gene ontology and pathway enrichment analysis

The WebGestalt 2019⁴⁸[\[48\]](#) online tool was used to perform the gene ontology and pathway enrichment analysis of the selected proteins. The biological process, cellular component and molecular function were all selected for gene ontology analysis and the Kyoto Encyclopedia of Genes and Genomes Pathways database was selected for pathway analysis. with FDR < 0.05 were considered significant. IPA analysis was conducted by QIAGEN Ingenuity Pathway Analysis software. The protein list was uploaded to the software and then annotated with the default settings.

Statistical analysis

All statistical analyses were conducted using R 4.2.1 software. Significance was determined by the Wilcoxon rank-sum test unless specified otherwise. Significance was concluded if the p-value was <0.05, while in pathway analysis, significance was concluded if the FDR was <0.05 after correction for multiple comparisons.

Disclosure of conflict of interest

UT MD Anderson Cancer Center and R.K. hold patents in the area of exosome biology licensed to Codiak Biosciences, Inc. UT MD Anderson Cancer Center and R.K. are stock equity holders in Codiak Biosciences, Inc. R.K. is a consultant and scientific adviser for Codiak Biosciences, Inc. The remaining authors declare no competing interests.

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Data availability

The public datasets used in this study were described in the Methods sections.

Contributions

B.L. and R.K. conceived and designed the study. B.L. developed the computational algorithm and performed the analysis with supervision from R.K. F.G.K. generated the proteomics data of the cell lines. B.L. and R.K. wrote the manuscript. All the authors read the manuscript and discussed the results.

Supplementary Figure S1. Machine learning models for plasma-derived exosomes. (A) PCA plot of cancer and control plasma or serum-derived exosomes. **(B)** Protein abundance of 18 selected protein features in cancer- and control plasma or serum-derived exosomes.

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Reviewer #1 (Public Review):

Summary:

In this manuscript, the authors used machine learning algorithm to analyze published exosome datasets to find biomarkers to differentiate exosomes of different origin. By applying the method to "exosomes" sample, the author discovered common exosome markers and cancer-type specific markers.

Strengths:

The performance of the algorithm are generally of good quality.

<https://doi.org/10.7554/eLife.90390.2.sa1>

Reviewer #2 (Public Review):

Summary:

This is a fine work on the development of computational approaches to detect cancer through exosomes. Exosomes are an emerging biomarker resource and have attracted considerable interests in the biomedical field. Kalluri and co-workers collected a large sample pool and

used random forest to identify a group of protein markers that are universal to exosomes and to cancer exosomes. The results are very exciting and not only added new knowledge in cancer research but also a new and advanced method to detect cancer. Data was presented very nicely and the manuscript was well written.

Strengths:

Identified new biomarkers for cancer diagnosis via exosomes.

Developed a new method to detect cancer noninvasively.

Results were presented nicely and manuscript were well written.

<https://doi.org/10.7554/eLife.90390.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1:

Summary:

In this manuscript, the authors used machine learning algorithm to analyze published exosome datasets to find biomarkers to differentiate exosomes of different origin.

Strengths:

The performance of the algorithm are generally of good quality.

Weaknesses:

The source datasets are heterogeneous as described in Figure 1 and Figure 2, or Line 72-75; and therefore questionable.

Response: We thank the reviewer for this assessment. The commonly used biomarkers of exosomes exhibit heterogeneous presence and abundance within the exosomes derived from different cell lines, tissue, and biological fluids. The primary goal of this study was to identify universal exosomal biomarkers that remain consistent across different sources of exosomes, unaffected by potential isolation and quantification bias. This objective was achieved through an integration of datasets from different sources, which allowed for the subsequent identification of common proteins associated with exosomes. Among the 18 protein markers identified, it is noteworthy that they are universally abundant in all cell lines and their exosomes. We believe that despite the heterogeneity of the datasets used here, the identification of 18 universal protein markers in exosomes from diverse sources is a strength of this analysis.

(1) Nomenclature: Extracellular vesicles (EVs) are small vesicles released by cells into the extracellular space, exhibiting high heterogeneity in origin across species. Exosomes are typically defined as being of multivesicular body origin. However, the absence of several crucial common exosomal markers, including CD63, suggests that the proteomics analysis may include various other vesicular and non-vesicular materials.

Response: As we reported previously (Kugeratski et al., Nature Cell Biology, 2021), the commonly used exosomal markers, such as CD9, CD63 and CD81 exhibit heterogeneity with respect to presence and abundance in the exosomes derived from different cell types. For example, CD63 demonstrated remarkably lower abundance in the exosomes derived from Raji cell lines. In our study, the detection rate of CD63 (< 50%) is quite low in the tissue-

derived exosomes, which is consistent with the observations made in another proteomics based study (Hoshino et al., Cell, 2020). Therefore, relying solely on these markers is inadequate for the comprehensive characterization of EVs as exosomes. Therefore, we conducted this study to identify universal protein markers of exosomes by integrating data from multiple sources, thereby circumventing potential confounding effects due to their diverse origins and other technical differences.

(2) Line 90: IPA is not prior in the manuscript.

Response: We provided the full definition of IPA (Ingenuity Pathway Analysis) in the revised manuscript.

(3) Figure 2B: Considering the large number of variables, it is unsurprising that the 2D PCA (Principal Component Analysis) falls short in the classification task. Including a few additional dimensions (principal components) might have the potential to better distinguish the cancer groups from the control group.

Response: Thank you for this insightful query. The purpose of utilizing PCA here is to appreciate the heterogeneity associated with exosomes from different studies. While we acknowledge that additional dimensions may be more useful in distinguishing between cancer and control exosomes, we believe that derived performance will remain inferior to the machine learning approach we developed here.

(4) Figure 2D: Exosomes primarily derive from multivesicular bodies, rather than the plasma membrane. It remains unclear why the authors focus specifically on proteins in the plasma membrane. Is it intended to encompass all membrane proteins? Clarification is needed on this point.

Response: A good point. This study attempted to identify protein biomarkers of exosomes originating from different sources. Our approach involved considering proteins present on the plasma membrane as potential biomarkers also because many of them have been detected on the surface of exosomes.

(5) Figure 2F: The 18 identified proteins are also abundantly present in control cells, not solely in cancer-derived "exosomes." The statement in line 104 is misleading in this regard.

Response: We apologize for the misleading sentence. We have revised the statement to state that "In total, we identified a set of 18 exosome protein markers that are present at a higher abundance in all exosomes examined".

(6) Figure 3B: Considering the definition of exosomes, CD63 and TSG101 should be present in all samples, and their absence raises concerns.

Response: We understand the concern of the reviewer. In this Figure, we analyzed CD63 and TSG101 in tissue-derived exosomes. Our results are consistent with the previous study also shows the paucity of these makers in the tissue-derived exosomes (Hoshino et al., Cell, 2020). Our study highlights that CD63 and TSG101 cannot always identify exosomes from diverse cell lines and tissues. Such initial observations motivated us to conduct this study to identify the universal biomarkers of exosomes across different sources.

(7) Figure 6G&H: Achieving an accuracy of 80% cannot be deemed "excellent."

Response: We employed the word “excellent” in line 225 to describe the sensitivity and specificity associated with AUROC.

(8) Other comments on methods: The manuscript lacks an explanation of the neural network structure and why it outperforms other methods. Additionally, details about the calculation of MI (mutual information), IPA, and other methods should be provided.

Response: This is a good suggestion but in this work we did not employ the neural networks for the analysis. We provided additional details and explanations regarding the methodology for mutual information score calculation, as well as insights into the improved use of IPA and other relevant methods in the revised manuscript.

Reviewer #2:

Summary:

This is a fine work on the development of computational approaches to detect cancer through exosomes. Exosomes are an emerging biomarker resource and have attracted considerable interests in the biomedical field. Kalluri and co-workers collected a large sample pool and used random forest to identify a group of protein markers that are universal to exosomes and to cancer exosomes. The results are very exciting and not only added new knowledge in cancer research but also a new and advanced method to detect cancer. Data was presented very nicely and the manuscript was well written.

Strengths:

Identified new biomarkers for cancer diagnosis via exosomes.

Developed a new method to detect cancer non-invasively.

Results were presented nicely and manuscript were well written.

Weaknesses:

N/A.

Response: We appreciate the the enthusiastic assessment of our study by the reviewer.

Reviewer #3:

In the current study, Li et al. address the difficulty in early non-invasive cancer diagnosis due to the limitations of current diagnostic methods in terms of sensitivity and specificity. The study brings attention to exosomes - membrane-bound nanovesicles secreted by cells, containing DNA, RNA, and proteins reflective of their originating cells. Given the prevalence of exosomes in various biological fluids, they offer potential as reliable biomarkers. Notably, the manuscript introduces a new computational approach, rooted in machine learning, to differentiate cancers by analyzing a set of proteins associated with exosomes. Utilizing exosome protein datasets from diverse sources, including cell lines, tissues, and various biological fluids, the study spotlights five proteins as predominant universal exosome biomarkers. Furthermore, it delineates three distinct panels of proteins that can discern cancer exosomes from non-cancerous ones and assist in cancer subtype classification using random forest models. Impressively, the models based on proteins from plasma, serum, or urine exosomes achieve AUROC scores above 0.91, outperforming other algorithms such as Support Vector Machine, K Nearest Neighbor Classifier, and Gaussian Naive Bayes. Overall, the study presents a promising protein biomarker signature tied to cancer exosomes and proposes a machine learning-

driven diagnostic method that could potentially revolutionize non-invasive cancer diagnosis.

Response: We appreciate this positive assessment of our work.

(1) The authors should clarify why they focused solely on protein markers. Why weren't RNA transcripts also considered? Do the authors see value in incorporating RNA/micro RNA transcripts to enhance diagnostic capabilities?"

Response: This is a very important point for further consideration. The current datasets for exosomal proteins are extensive and generally proteins might offer distinct advantages in cancer diagnostics compared to nucleic acids due to their stability in exosomes and extended half-life (Schey et al., Methods, 2015). We do agree that the power of analysis can only get better if also add DNA, RNAs and other constituents and we hope to pursue such analysis in the future.

(2) Can the identified exosomal markers also be evaluated as prognostic indicators?

Response: We appreciate this intriguing question. Indeed, proteins such as apolipoprotein E (APOE) may serve as a potential prognostic marker in various cancers (Ren et al., Cancer Medicine, 2019). APOE is being extensively studied as a prognostic and diagnostic marker for multiple cancer types, including colorectal cancer (Martin et al., BMC Cancer, 2014), gastric cancer (Sakashita et al., Oncology Reports, 2008), pancreatic cancer (Chen et al., Medical Oncology, 2013; Xu et al., Tumor Biology, 2016), and human hepatocellular carcinoma (Yokoyama et al., International Journal of Oncology, 2006). In these studies, APOE levels were found to be elevated in the serum of cancer patients and correlated with survival outcomes.

(3) The discussion should emphasize if the identified protein markers are tumor-specific or if they indicate, for instance, the patient's immune reaction to the tumor.

Response: A good point. We believe that the identified biomarkers are tumor-specific and a significant number of these proteins have been previously associated with tumor initiation and progression. Further studies will likely identify immune response-related biomarkers when more in-depth tumor-level analyses are performed.

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